

Master Build Instructions: AI Fitness Coach

1. Project Overview

The **AI Fitness Coach** is an AI-powered platform delivering personalized diet and workout plans using body analysis, user goals, and continuous tracking. It is extended with a robust Admin Panel for monitoring, control, and optimization.

2. Core Technology Stack

Layer	Technologies	Purpose
Frontend	React, Tailwind CSS / Shadcn / Material UI	Build separate User and Admin dashboards, Chat UI, and Image upload interfaces.
Backend	Node.js + Express	Handle REST APIs and secure routing.
Database	MongoDB	Manage strict collections via NoSQL structure.
Real-Time	Socket.IO	Power instant chat updates and live tracking sync.
AI / Vision	OpenAI / Gemini, MediaPipe, RAG System	Generate plans, estimate BMI/posture, and power context-aware chat.

3. Security & Authentication

- **Auth:** Implement JSON Web Tokens (JWT) for secure authentication.
- **RBAC:** Enforce strict Role-Based Access Control separating user and admin roles.
- **Middleware:** Utilize specific middleware functions: verifyJWT and checkRole('admin').

4. Database Collections (MongoDB)

Collection Name	Stored Data Types
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Users	User accounts categorized by role (user or admin).
Plans	AI-generated diet and workout routines (deeply nested JSON).
Progress	Daily habit tracking data and weekly photo updates.
Chat logs	RAG chatbot conversation history.
Admin logs	System logs, AI usage logs, and error tracking.

5. End User Core Features

- **AI Onboarding:** Process 4 uploaded images (Front, Back, Left, Right).
- **Computer Vision:** Use MediaPipe for posture detection, body landmarks, and estimated BMI.
- **Goal Selection:** Capture user targets (Weight Loss, Gain, Muscle, or Maintenance).
- **Plan Generation:** Create allergy-aware diet plans with macros/calories and detailed workout plans (sets, reps, weekly splits).
- **Daily Tracker:** Monitor meals, water, workouts, and sleep utilizing a streak system.
- **AI Chatbot (RAG):** Deliver context-aware answers based on the user's specific plan and progress history.
- **Weekly Progress:** Facilitate photo comparisons and provide AI insights.

6. Admin Panel Features

- **Analytics Dashboard:** Monitor total/active users, plan completion rates, chat usage, and average fitness scores.
- **AI Output Monitoring:** View generated plans, flag inaccurate outputs, and manually adjust AI prompt templates.
- **Plan Override:** Edit AI routines, create custom templates, and assign manual plans directly to users.
- **Moderation:** Review and delete uploaded body images, detect harmful chat queries, and block abusive users.

7. System Constraints & Safety Warnings (Strict Limits)

- Acknowledge that the AI is **not medically certified**.
- Ensure all image-based body analysis is clearly labeled as **approximate**.
- Enforce active moderation as a strict requirement for overall safety.